

High Energy-efficiency and Low latency In-Memory Computing using Analog Accumulator and In-Memory ADC with shared References

Junyi Yang, Shuai Dong, Zhengnan Fu, Hongyang Shang, Arindam Basu

Department of Electrical Engineering, City University of Hong Kong, Hong Kong, China

Abstract—This article proposes a 256×128 in-memory computing array using reconfigurable in-memory analog-to-digital conversion with shared references for high area efficiency (area overhead of 3% is 9X better than traditional). A dual-8T SRAM bitcell is used to achieve read-write decoupling and store ternary weights. Read World Line Under Drive enabled Cascode helps to minimize current variations producing high linearity. Multi-bit input is handled with low latency and high energy-efficiency by using bit-slicing (BS) with near-memory charge-sharing based binary weighted accumulator (CHA). Using noise resilient training, we show software comparable performance for a MLP on MNIST, VGG-8 on CIFAR-10, and graph attention network on Cora, with respective accuracy reductions of only 0.1%, 0.8% and 0.5% due to non-idealities. The proposed macro demonstrates high energy/area efficiency (1146 TOPS/W, 27 TOPS/mm² at 1/2/1b) in 65 nm CMOS. It increases throughput (by 1.9X) and linearity (by 23X) compared to input pulse-width modulation by using BS and CHA. Compared to conventional BS with digital accumulation after ADC, this method has 1.7X/6.6X better energy-efficiency/throughput by reducing ADC operations.

Index Terms—In-memory computing, In-memory ADC, SRAM, Charge-sharing, Multi-bit

I. INTRODUCTION

Deep Neural Networks (DNNs) produce state-of-the-art solutions across different domains such as speech processing [1], image classification [2], natural language processing [3] etc. However, the bottleneck in implementing DNNs on current hardware stems from the frequent memory access requirements imposed by the von Neumann architecture, which is known as the memory wall or von Neumann bottleneck [4], [5]. Moreover, as DNN models expand in scale, the heightened energy consumption and latency associated with storing the parameters of sizable models further exacerbate this challenge.

In-memory computing (IMC) has emerged as a promising approach to address these limitations [6]. IMC involves conducting computations within the memory itself, specifically on its bit line, rather than repeatedly reading and writing data from memory during each cycle. Furthermore, IMC enables the execution of multiply-and-accumulate (MAC) operations

within each column by activating multiple rows, thereby significantly diminishing the generation of intermediate data and enabling highly parallelized computation [7]. Among the various IMC techniques that have been investigated, analog/mixed-signal IMC utilizing SRAM has shown substantial enhancements in terms of latency, energy/area efficiency, and throughput [8], [9]. However, several challenges remain to be solved such as resilience to PVT (process-voltage-temperature) variations, reducing energy and area overhead of peripheral data converters, reducing non-linearity and enabling high bit-precision MAC operations with low-latency, energy and area overhead.

In this work, we propose a multi-bit, highly reconfigurable IMC (1-7/2-4/1-7b, input/weight/output) by using bit-slicing (BS) with charge-sharing based binary weighted accumulator (CHA) and reconfigurable (1-7b) In-memory Analog to Digital Converter (IMADC) with shared references. A dual 8T SRAM bitcell based on Read World Line Under Drive Cascode (RWLUDC) has been employed in the design of a macro to minimize current variations with bitline voltage change enabling larger signal margin. Our work results in 1.9X and 6.6X improvement in latency over conventional pulse width modulation (PWM) mode and BS mode respectively under 7-bit input/output by using bit-slicing with charge-sharing based binary weighted accumulator (BSCHA) as shown in Fig. 1(a). It also increases MAC linearity by $\approx 23X$ compared to PWM mode. IMADC with shared references diminishes the hardware overhead without reducing throughput. Our ADC area overhead (ADC area/MAC array area) is merely 3%, which is much lesser than prior work in Fig. 1(b) (9X improvement compared to 27% for traditional IMADC). We also show inference accuracies at par with software baselines due to noise aware training and show requirement for high input/output bit resolution (7b) for complex networks like graph attention.

II. BACKGROUND AND RELATED WORK

A major issue limiting the application of analog IMC is the area, energy and latency overhead of ADC [10]. To mitigate the area overhead of traditional ADCs [11], [12], column multiplexing is used where M columns share one ADC. However, this comes at the cost of reduction in system throughput by a factor of M . Therefore, IMADC was proposed [13], [14], where each column can operate a slow ramp ADC simultaneously. This has similar or better throughput as earlier

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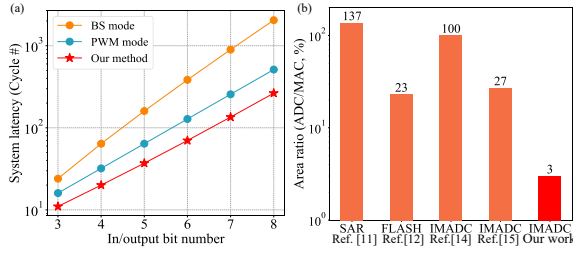


Fig. 1. System latency and ADC area overhead: (a) System latency (Our method, PWM mode and BS mode). (b) ADC area overhead (ADC area/MAC array area).

designs [11], [12] but with the advantage of reconfigurability (of resolution and MAC size) as well as robustness to PVT variations due to replica biasing. However, only binary NNs could be implemented, which fall short of meeting the accuracy demands of intricate neural networks [13], [14]. Also, integrating an ADC into each column significantly increases the hardware overhead, especially as the resolution of the ADC, n_o , increases. For instance, achieving a 5-bit ADC requires 64 extra bit cells in every column. Fig. 1(b) shows ADC area overhead defined as the ratio of ADC area over MAC array area in previous works [11], [12], [14], [15].

BS is a popular approach for increasing the input bit resolution (n_i), where input bits are applied serially and partial sums (psums) are combined after digitization. However, when employing a ramp ADC for output quantization, the total latency of BS required becomes $n_i \times 2^{n_o}$ clock cycles, since each bit demands 2^{n_o} clocks. Also, this incurs n_i times energy overhead of ADC as well as additional digital circuits for combining psums. Another alternative, PWM for multi-bit input [16]–[18] incurs a time cost of $2^{n_i} + 2^{n_o}$, which is also notably high. PWM also requires Digital-Analog converters (DAC) at the input or counters with very fast clocks. Fig. 1(a) illustrates this tradeoff between system latency and input/output bit resolution (assuming $n_i = n_o$).

III. PROPOSED ARCHITECTURE AND METHODOLOGY

A. Overall Hardware Architecture

Fig. 2(a) shows the top-level schematic of the IMC and IMADC based on SRAM, which includes 256×127 dual 8T SRAM bitcells for MAC, one column (256×1) for references of the reconfigurable ADC (shared by 127-column MAC array), charge-sharing circuits, sense amplifiers(SA), ripple counters, input drivers and voltage buffers. To clearly illustrate how it works, the timing of MAC operations and a set of ADC timing diagrams are depicted in Fig. 2(b). Input bits b_i are applied serially to create $V_{MAC} = V_{MACP} - V_{MACN} = \sum_{k=1}^N W_k X_k^i$ on the bitlines (using current-mode operation) where $N = 256$ is the dimension of the input vector. The near memory capacitors $C_{X1} = C_{X2}$ perform binary weighted accumulation of partial sums in charge mode to create the final accumulated voltage V_{Acc} . Next, the ADC column creates a reference ramp voltage ΔV_{ADC} that increases every clock cycle. This is shared with the remaining columns so that each

of the 127 SAs can simultaneously compare the ramp voltage with their corresponding V_{Acc} values. Due to the shared nature of the ramp voltage, differential voltage buffers are required to enhance the driving capability of the ramp reference voltage. The ripple counter converts the output thermometer code of the SAs into a binary code. Details of each block are described next.

B. Dual 8T SRAM based In-Memory Binary Multiplication

A dual 8T SRAM bitcell has been utilized for the implementation of the macro as shown in Fig. 2(c). It performs binary multiplication of an input with a ternary weight using a decoupled read path, which includes the four blue NMOS transistors (two transistors on each side as shown in Fig. 2(c)). The read wordline (RWL) receives either a 1 (a high DC voltage) or a 0 (a low DC voltage) signal according to the input bit, while a ternary weight (-1: Q_L is low and Q_R is high, 0: both Q_L and Q_R are low, +1: Q_L is high and Q_R is low) is stored in the SRAM bitcell. Initially, both RBLL and RBLR voltages are pre-charged (by signal PCH_MAC in Fig. 2(b)) to a high voltage using two pre-charge PMOS transistors before the binary multiplication operation commences. A table in Fig. 2(c) illustrates the multiplication results of the bitcell. When the input is 0 (RWL is low), no current discharges the RBLL or RBLR, regardless of the weight stored in the SRAM bitcell. Conversely, When the input is 1 (RWL is high), either RBLL or RBLR is discharged via the corresponding NMOS transistor, where the gate is linked to the internal 6T-SRAM cell with a high voltage (0.5 V). The resulting voltage disparity between RBLL and RBLR ($+\Delta V$ or $-\Delta V$) signifies the binary multiplication outcome. The layout of the dual 8T cell shown in Fig. 2(c) occupies $3.6 \mu\text{m} \times 1.8 \mu\text{m}$.

Accumulation occurs due to all bitcells in a column simultaneously discharging the bitline capacitor C_{BL} . The MAC value is directly proportional to the voltage drop induced by the capacitor discharge process. A more detailed explanation of this process will be provided in Section III-D.

C. RWL Underdrive enabled Cascode for enhanced linearity

A major problem of current based MAC architecture is maintaining linearity of C_{BL} discharge due to variation in discharge current of a cell, I_u , with change in V_{DS} as the voltage on BL changes [13], [14]. This can be improved by increasing the output impedance of the effective current source created by transistors M0(L/R) and M1(L/R) in Fig. 2(c) by making M1 behave as a cascode [19]. We propose to achieve this by RWL underdrive voltage enabled cascode (RWLUDC) technique—the supply voltages for RWL is set to 0.8 V while RBLL/RBLR pre-charging are set at 1 V while gate voltage of M0 is at 0.5 V (SRAM supply voltage). The minimum RBL voltage for which M1 acts as a cascode is given by $V_{min} = V_{RWL} - V_{T1}$ where V_{T1} denotes the threshold voltage of M1, i.e. lower V_{RWL} leads to higher RBL swing. As a result, we achieve a voltage swing range of 700 mV on the RBL for a 1% change in I_u (Fig. 2(c))—7X less than the variation in I_u over the same range for the conventional case.

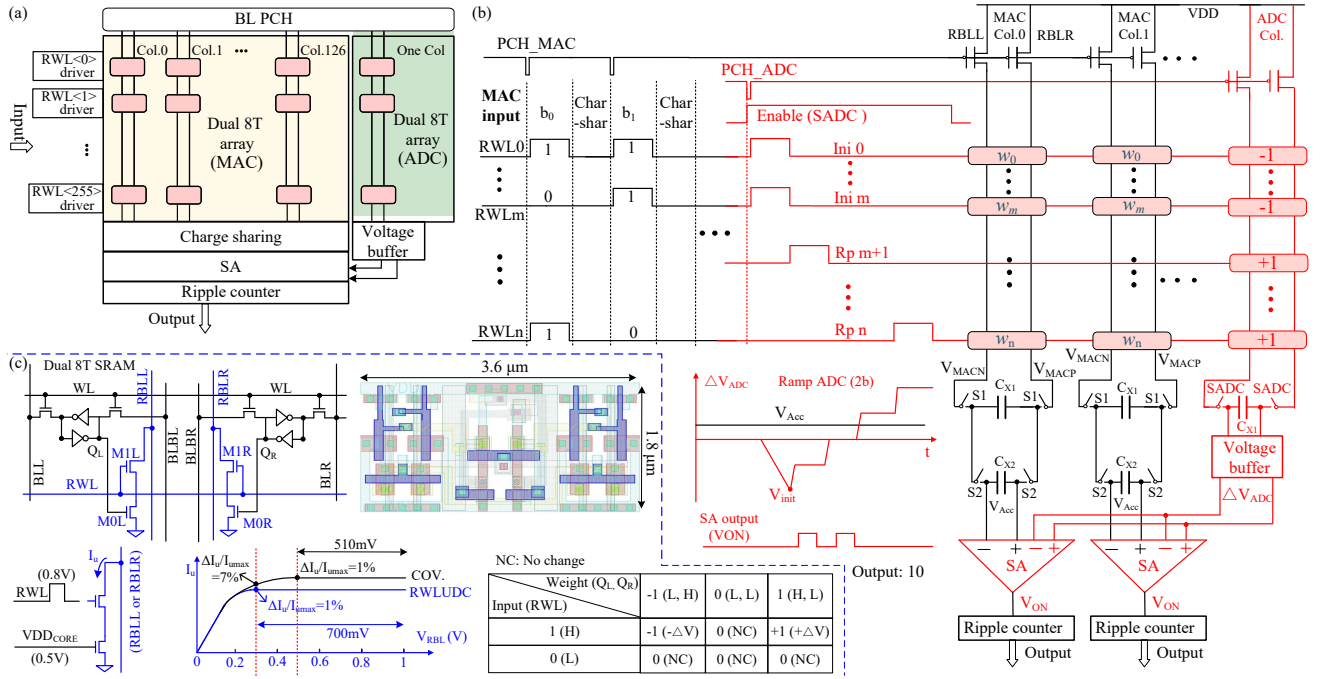


Fig. 2. Overall structure of the proposed dual 8T SRAM IMC and IMADC: (a) system hardware structure. (b) Detailed circuit and timing diagram of MAC columns and IMADC. (c) Information of dual 8T SRAM cell.

Alternatively, the conventional method needs to reduce signal swing by $1.4X$ to ≈ 510 mV for the same 1% variation. Single transistor (in 7T cell) based discharge paths have an even smaller signal swing range of ≈ 200 mV [15].

D. Bit Slicing with Charge Sharing based Binary Weighted Accumulator (BSCHA) for Multi-bit Input

Conventional BS in IMC applies the input bits b_0 to b_{n_i-1} serially over n_i cycles while the MAC voltage is digitized separately for every bit to produce psums P_0 to P_{n_i-1} . The final MAC result is obtained by weighted summation in the digital domain as $P_{n_i-1} + \frac{P_{n_i-2}}{2} + \dots + \frac{P_0}{2^{n_i-1}}$. The resulting energy (E_{conv}) and latency (T_{conv}) in terms of clock cycles of the conventional BS approach are given by:

$$\begin{aligned} E_{conv} &= n_i(E_{MAC} + E_{ADC} + E_{dig}) \\ T_{conv} &= n_i(T_{MAC} + T_{ADC} + T_{dig}) \end{aligned} \quad (1)$$

where $E(T)_{MAC}$, $E(T)_{ADC}$ and $E(T)_{dig}$ denote the energy (latency) of MAC, ADC and digital weighted accumulation respectively for every cycle. This incurs a heavy penalty in terms of energy and latency due to the repeated operation of the ADC over n_i cycles.

To overcome this issue, we propose a binary weighted accumulation before digitization technique accomplished by the addition of two capacitors and four switches between RBL and RBLR (Fig. 2(b)). The simplified circuit diagram is illustrated in Fig. 3(a), where C_{BL} represents the parasitic capacitance on RBL, and I_{MACN} and I_{MACP} are the total discharge currents of each column. Intuitively, C_{X1} is used to sample the MAC voltage in every cycle for every input

bit, and then redistribute it with an **equally sized** capacitor C_{X2} . This process results in the binary weighted charge sharing based accumulation (CHA) required to combine the partial sums for every bit. It is similar in principle to charge redistribution DACs [19]; however, instead of an input digital code, a different analog MAC voltage is input in every cycle.

The detailed process is shown in Fig. 3(b) including the timing diagram of pre-charging, input pulse, and charge-sharing control switches S1 and S2. We analyze the case for the i -th bit of the input (assuming both capacitors were reset before the first input bit was presented):

1) During pre-charging, Switch S1 is closed, and S2 is open. After pre-charging, the voltages of the two C_{BL} s and the voltages across C_{X1} reach VDD resetting the previously stored voltages.

2) Subsequently, RWL is enabled for time Δt , initiating the discharge of C_{BL} . The voltage on C_{X1} changes as the two C_{BL} s discharge. By applying Kirchhoff's Current Law, we can derive Eq. (2) for the voltage $V_{MAC} = V_{MACP} - V_{MACN}$ as:

$$V_{MAC}^i = \frac{(I_{MACP}^i - I_{MACN}^i) \Delta t}{2C_{X1} + C_{BL}} = \frac{I_u \sum_{k=1}^N W_k X_k^i \Delta t}{2C_{X1} + C_{BL}} \quad (2)$$

where V_{MAC}^i denotes the MAC voltage developed on the capacitors C_{BL} and C_{X1} when the i -th input bit is presented to the circuit. The term $2C_{X1}$ can be understood from Miller's effect applied to a floating capacitor [19].

3) Subsequently, S1 opens, and S2 closes, initiating the charge-sharing between capacitors C_{X1} and C_{X2} . The voltage on capacitor C_{X2} denoting the final accumulated voltage V_{Acc} becomes a combination of half of its previous value V_{Acc}^{i-1} and

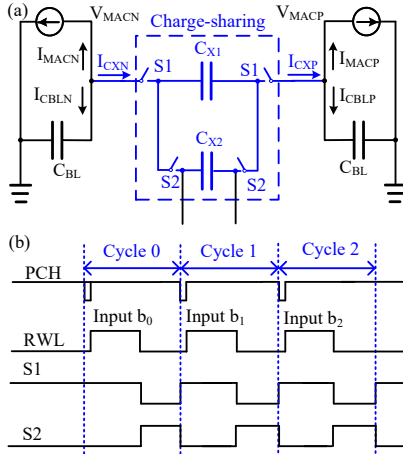


Fig. 3. (a) Effective circuit model for generation of differential MAC voltage and charge sharing. (b) Timing diagram of charge sharing. The MAC voltage is sampled on C_{X1} when $S1$ and RWL are enabled, while charge redistribution between C_{X1} and C_{X2} occurs when $S2$ is enabled.

half of the the new MAC value V_{MAC}^i as shown in in Eq. (3):

$$\begin{cases} C_{X2}V_{Acc}^{i-1} + C_{X1}V_{MAC}^i = V_{Acc}^i(C_{X1} + C_{X2}) \\ V_{Acc}^i = \frac{V_{Acc}^{i-1}}{2} + \frac{V_{MAC}^i}{2} = \sum_{k=0}^i \frac{V_{MAC}^k}{2^{i+1-k}} \end{cases} \quad (3)$$

Compared to other charge sharing methods [11], our proposed method does not require binary weighted capacitors, reducing area and matching requirements. In our implementation, $C_{X1} = C_{X2} = 50$ fF is chosen to make it approximately similar to $C_{BL}/2$ (balancing two parts in denominator of Eq. (2)) since it provides a good tradeoff between reducing noise and maintaining large enough MAC voltages. The cell array layout is designed to utilize metal layers M1 to M4. To optimize the overall chip area, metal layers M6 to M7 are allocated for constructing MOM capacitors C_{X1} and C_{X2} , while metal layer M5 is grounded to shield the cell array from the capacitors. Each capacitor has an area of $3 \times 47 \mu m^2$, enabling the straightforward placement of two MOM capacitors on top of every column in the cell array without incurring any additional area overhead.

The energy and latency for our approach can be written as:

$$\begin{aligned} E_{proposed} &= n_i(E_{MAC} + E_{ana}) + E_{ADC} \\ T_{proposed} &= n_i(T_{MAC} + T_{ana}) + T_{ADC} \end{aligned} \quad (4)$$

where $E(T)_{ana}$ denotes the energy (latency) of analog weighted accumulation and other terms are as defined earlier. The major advantage of our approach stems from the fact that the ADC is now operated only once as opposed to n_i times as in Eq. (1). The errors induced by the analog accumulation can be compensated by co-design of the algorithm as shown in Sec. III-F.

In contrast to the PWM mode where the input latency is 2^{n_i} , our method demonstrates clear advantages in terms of input latency, as shown in Fig. 1(a) (18X improvement for

7-bit input). Furthermore, in comparison to the PWM mode for multi-bit input, our method exhibits superior linearity in generation of V_{Acc} due to smaller voltage swing on V_{RBLL}/R . As mentioned in Section III-C, Early effect of discharge transistor of the bit cell ($M0$, $M1$) results in a decrease in I_u , subsequently diminishing V_{Acc} and giving rise to considerable errors.

E. Reconfigurable 1-7 bit In-Memory Analog-Digital Converter (IMADC) with Shared References

The red part in Fig. 2(b) shows the architecture and operation of IMADC. The replica cells for ADC are exactly the same as the MAC part. Conceptually the operation is split into two parts: (1) generation of initial ramp voltage V_{init} (Fig. 2(b) bottom) in one clock cycle by first enabling 2^{n_o-1} bit-cells with weight of -1 , and (2) generation of ramp voltage by enabling 2^{n_o} bit-cells with weight of $+1$, one cell enabled every clock followed by SA based comparison each time. Fig. 2(b) depicts the schematic outline of the global ramp reference voltage (ΔV_{ADC}) generation for a 2-bit ADC and the quantization process of the voltage in the MAC column (V_{Acc}).

Earlier work [14] used ramp generators in every column leading to $\approx 50\%$ overhead in terms of area. In the proposed architecture, we replace it with a global reference generator. Due to shared RWL lines between MAC and ADC ramp generator, an *Enable* signal ($SADC$) is used to ensure the ADC starts after the MAC operation ends, thus saving switching and voltage buffer power dissipation. $SADC$ is used to gate the PCH and RWL to create PCH_ADC and RP signals for the ADC column. It also enables the SA. Depending on the required resolution, different number of cells can be used for the ramp generation. A maximum resolution of $n_o = 7$ can be obtained with the $N = 256$ rows in the macro; this uses 192 cells while the remaining 64 can be used for calibration to ensure the ramp crosses “0” at the mid-code [14].

While the proposed method reduces area overhead to $\approx 3\%$ (Fig. 1(b)), it requires two modifications—(1) the SA has to be converted to a double differential one [20] that compares the differential ramp voltage ΔV_{ADC} with a differential V_{Acc} and (2) a voltage buffer has to be used after the ramp generator to drive the input capacitance of 127 SAs. The output signal V_{ON} of the SA is the input of the ripple counter. In the scenario where $V_{Acc} = 2$, as illustrated in Fig. 2(b) (bottom), the output of the ripple counter is 10. The ripple counter acts as a thermometer to binary code converter, eliminating the need for large register arrays to store the thermometer code, thereby enhancing the area efficiency of the system.

Two self-biased voltage buffers is utilized in the IMADC (Fig. 4(a)). They are designed to operate at a frequency of 200 MHz with load capacitance of 100 fF while dissipating 40 μW from a power supply of 1 V. In the Monte Carlo (MC) simulation depicted in Fig. 4(b), the offset voltage determined by the output difference of the two voltage buffers minus the input difference is shown. The MC results indicate that the mean offset is 3.3 mV, with a standard deviation of 0.1 mV.

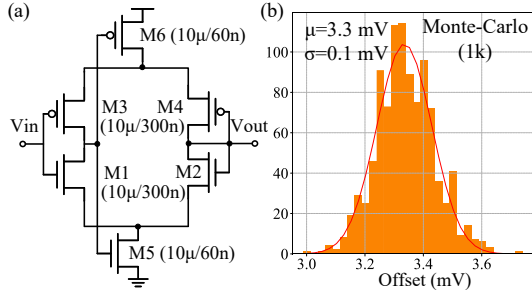


Fig. 4. Self-biased voltage buffer (a) Circuit Diagram. (b) Monte Carlo simulation for voltage offset showing it to be less than ADC LSB.

Algorithm 1 Noise resilient training.

Input: X_t , W_t and σ
Output: W_{t+1}
loop
 Feed forward propagation:
 $Y_t \leftarrow Q(W_t \cdot X_t)$
 $Z_t \leftarrow f(Y_t + \sigma)$
 Back-propagation propagation:
 $g_t \leftarrow$ calculate gradient based on W_t and $f(W_t \cdot X_t)$
 $W_{t+1} \leftarrow W_t + \eta \times g_t$
end loop

This is smaller than the ADC LSB of 4.8 mV and removes the need for calibration. Even with the added buffers and more complex SA, the proposed method dissipates $\approx 3\%$ less energy than having a ramp generator in every column [13], [14] due to the reduced bitline capacitances and switching energy of our shared reference scheme.

F. Noise-resilient Training (NRT)

To mitigate the impact of non-idealities (such as mismatch between C_{X1} and C_{X2} , discharging currents of the bitcells etc.) in the proposed architecture, noise-resilient training (NRT) is used to compensate for the accuracy loss, inspired by quantization-aware training. Details of the method are shown in Algorithm 1 where X_t , W_t , Y_t , Z_t and g_t denote the input, weight, ideal quantized MAC value, non-ideal activation value and gradient on the t -th iteration. $Q()$ and $f()$ are the ADC quantization and activation functions. σ denotes samples from a noise distribution mimicking the hardware non-idealities and is obtained from circuit simulation (Fig. 5(a)). As shown in Algorithm 1, in the feed-forward propagation stage, ADC output with random noise is sent to the activation function. However, in back-propagation, ideal MAC values without noise are used to calculate the more accurate gradient and then update the weight.

IV. SIMULATION RESULT

A. Software results

We choose three networks to evaluate the proposed method: one MLP network (784-128-128-10) on MNIST dataset, VGG-

8 on CIFAR-10 dataset, and graph attention (GAT) network [21] on Cora dataset.

To conduct noise-resilient training (NRT), the proposed architecture is evaluated by using SPICE simulation in 65 nm CMOS technology. In Fig. 5(a), we compare 4-bit ADC outputs with the theoretical MAC values at different temperatures (0°C , 27°C and 70°C) and the extracted errors approximately follow normal distributions. It can be seen that the error is always small, thereby indicating the robustness of the IMADC architecture in the face of temperature fluctuations due to replica biasing. Then the nominal error distribution ($N(-0.12, 0.83)$ at 27°C) is used to inject noise into the three networks. 1) the MLP has less than 0.2% accuracy decrease for 2-4 bit ADC compared to the quantized models due to the usage of NRT and shallow architecture of the model, which prevents deep accumulation of noise. 2) VGG-8 is a challenging model with deeper architecture. However, NRT can still restore precision to a comparable level of accuracy, with a decrease of less than 0.9% compared to 3-5 bit quantized model. 3) GAT is a novel graph neural network that requires a higher n_o for good performance. NRT demonstrate the ability to restore accuracy to within 0.9% for this model. The highest ADC bit resolution n_o used for these three models results in 0.8%, 1.2% and 0.7% drop in accuracy from their floating point baselines (98.0%, 93.4% and 84.0%), making them comparable to SOTA accuracy for quantized models [13], [21], [22].

B. Hardware results

Our hardware architecture supports multi-bit weights to cater to the requirements of complex networks. Multiple-bit weights can be realized by connecting multiple cells in parallel. For instance, in a 4-bit configuration, excluding the sign bit, the remaining three bits can be implemented using 1, 2, or 4 cells in parallel (for a total of 7 cells for one 4-bit weight). The sign of the weight can be intrinsically implemented using a dual 8T architecture based on usage of the left or right cells.

Latency and throughput: We assessed the latency of the system across three modes: proposed, PWM, and conventional BS, with respective latencies of $n+2^n$, 2^{n+1} , and $n2^n$, assuming the utilization of ramp ADC output and $n_i = n_o = n$. Fig. 1(a) illustrates that our work yields 1.9X and 6.6X improvement of throughput over PWM mode and BS mode under 7-bit input/output resolution.

Enhanced linearity of MAC: During on-chip inference, the MAC ($MAC_P - MAC_N$) in every column is computed separately as MAC_P ($MAC_P = \sum_{k=1}^N w_k x_k \quad (w_k \geq 0)$) and MAC_N ($MAC_N = \sum_{k=1}^N w_k x_k \quad (w_k < 0)$), and subsequently input to SA as a differential signal. To determine the maximum voltage drop across RBL in on-chip inference, the distribution of MAC_P and MAC_N across all mini-batches in the test set for MNIST is used for the proposed and PWM method for $n_i = 3$. In Fig. 6 (top), the distribution of MAC_P in the worst-case scenario is depicted, while Fig. 6 (bottom) illustrates the corresponding RBL simulated voltage values,

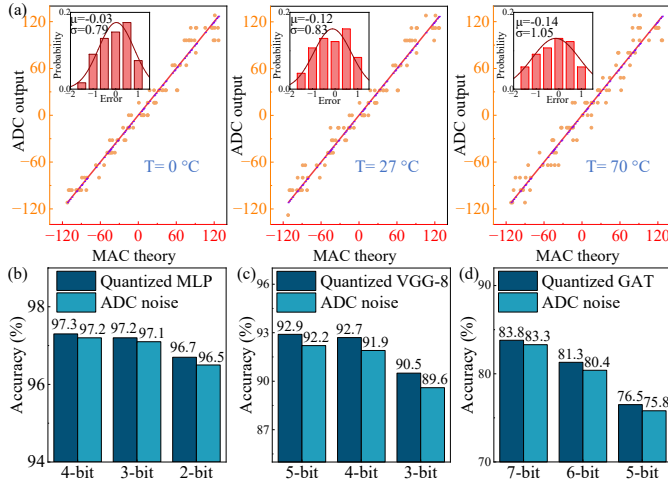


Fig. 5. Simulated results: (a) ADC output vs. theoretical result of MAC at 0°C, 27°C and 70°C. Test results for (b) MLP, (c) VGG-8 and (d) GAT.

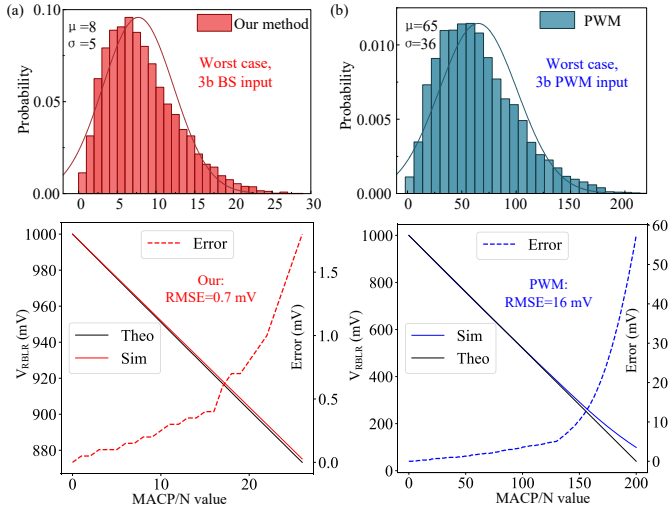


Fig. 6. The partial MACP/N distribution based on 3-bit input for MNIST and its corresponding voltage on RBL: (a) Proposed method. (b) PWM mode.

the theoretical values, and the error between them. Notably, the MACP/N distribution range in the PWM method is 7X greater than that in the our method reducing signal margin [18]. Furthermore, our method demonstrates a RBL voltage linearity that is $\approx 23X$ greater than that achieved by the PWM mode, as quantified by the Root Mean Squared Error (RMSE).

Energy and Area efficiency: Fig. 7(a) shows simulated energy breakdown of our work when $n_i = 4$ and weight is 2-bit. Pre-charging and SAs consume most of the energy and the energy consumed by the added voltage buffer is negligible, accounting for less than 2%. The layout of core area is depicted in Fig. 7(b). The core size is 0.24 mm² and 127 IMADCs only occupy only 3% of the size of MAC array (8X improvement compared to 23% in the best work [12]). Table I summarizes a performance comparison with the state-of-the-art analog IMC. The proposed macro achieves the best combination of energy/area efficiency (1146 TOPS/W (1.7X

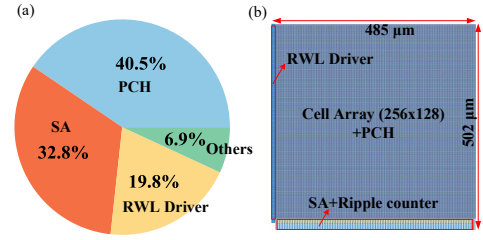


Fig. 7. (a) Energy breakdown (4-bit input/output and 2-bit weight). (b) Core layout of the macro.

TABLE I
COMPARISON WITH STATE-OF-THE-ARTS

	TCAS [†] 24 [15]	DAC [†] 21 [11]	JSSC [†] 20 [12]	Prop.
Tech. (nm)	65	28	65	65
Supp. (V)	0.45/0.8	0.5-1.1	0.6/0.8/1.0	0.5/0.8/1.0
Freq. (MHz)	200	150	50	200
Bitcell	Dual 7T	8T	8TIC	Dual 8T
Accum.	Current	Charge	Charge	Current+Charge
Array size	528x128	256x64	256x64	256x128
I/O bit #	1/1-6	2/5-6	1/5	1-7/1-7
Wgt. bit #	1.6/2.8/3.9	8	1	2-4
ADC type	Ramp (IMADC)	SAR	Flash	Ramp (IMADC)
ADC ovhd.*	100%	137%	23%	3%
Neuro # per ADC	1	8	1	1
GOPS	300-3900	21	1638	14-6502
TOPS/mm ²	0.9-10.4	0.75	20.2	0.06-27
TOPS/W	23.9-259	8.3-36	671.5	11-1146

* ADC ovhd.=ADC area/MAC area

than the best one), 27 TOPS/mm² (1.4X than the best one) at 1/2/1b and 14 TOPS/W, 0.06 TOPS/mm² at 7/4/7b) among the listed works and other recent publications [13], [23]. It is noteworthy that the proposed macro offers the highest level of reconfigurability in terms of precision for input (1-7b), weight (2-4b), and output (1-7b).

V. CONCLUSION

This article addresses important issues of IMC: multi-bit input, high linearity of MAC and large overhead of ADC. A dual 8T RWLU DC-based SRAM bitcell has been used to minimize current variations with bitline voltage change enabling larger signal margin. We enable multi-bit (1-7) input using BS with Charge-based accumulation of partial sums and multi-bit (1-7) output using a reconfigurable IMADC that reduces area overhead by shared references. The 256x128 in-memory computing array can achieve high efficiency (1146 TOPS/W, 27 TOPS/mm² at 1/2/1b for in/weight/out). Using NRT, it suffers minimal accuracy reductions of 0.1%, 0.8% and 0.5% due to circuit non-idealities for the MLP model (3/2/3b) on MNIST, VGG-8 model (4/2/4b) on CIFAR-10 and GAT network (7/3/7b) on Cora respectively. It increases throughput (1.9X) and linearity (23X) compared to PWM mode by BSCHA. Compared to conventional BS with digital accumulation after ADC, this method has 1.4X/1.7X/6.6X better area-efficiency/energy-efficiency/throughput respectively.

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